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PUBLICATION-READY REPORT (Paper Version)
Geotechnical ML Prediction Tool

Model Mapping Summary:
  PHI: Paper=Gradient Boosting | Tool=Gradient Boosting
  GAMMA_WET: Paper=XGBoost | Tool=XGBoost
  GAMMA_DRY: Paper=XGBoost | Tool=XGBoost
  SP_GR: Paper=Extra Trees | Tool=Extra Trees
  COHESION: Paper=Decision Tree | Tool=Gradient Boosting
  MC: Paper=Gradient Boosting | Tool=Gradient Boosting

BENCHMARK RATIONALE
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- Eleven representative machine learning algorithms were selected to span four major regression families: linear (Linear Regression, Ridge, Lasso), kernel-based (Support Vector Regression), a single decision tree, bagging ensembles (Random Forest, Extra Trees), and boosting ensembles (Gradient Boosting, XGBoost, LightGBM, CatBoost). These families were chosen because prior geotechnical machine-learning studies have consistently reported superior performance of ensemble and boosting methods for nonlinear SPT-based soil property prediction. All eleven algorithms were benchmarked under an identical borehole-group-aware cross-validation protocol to verify whether this trend held for the present Dinajpur dataset. Consistent with the literature, ensemble and boosting algorithms generally outperformed linear and kernel-based methods; the highest-performing algorithm for each target property was therefore selected for deployment in the final prediction tool. The full eleven-model benchmark is provided as supplementary material.

ENGINEERING INTERPRETATION
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- SP_GR was the most difficult property to predict (mean R2 = 0.387).
- COHESION was the most reliably predicted property (mean R2 = 0.809).
- For phi, Gradient Boosting gave the best fit (R2 = 0.843, RMSE = 2.658), attributed to its sequential error-correction mechanism, which captures nonlinear dependencies between SPT-derived predictors and soil properties.
- For gamma_wet, XGBoost gave the best fit (R2 = 0.645, RMSE = 1.487), attributed to its regularized gradient-boosting formulation, which controls overfitting while still capturing nonlinear soil-property relationships.
- For gamma_dry, XGBoost gave the best fit (R2 = 0.586, RMSE = 0.984), attributed to its regularized gradient-boosting formulation, which controls overfitting while still capturing nonlinear soil-property relationships.
- For sp_gr, Extra Trees gave the best fit (R2 = 0.460, RMSE = 0.111), attributed to its extremely randomized splitting rule, which further reduces variance and produced more consistent performance across different soil-property targets.
- For cohesion, Decision Tree gave the best fit (R2 = 0.963, RMSE = 1.386), attributed to its ability to capture simple nonlinear thresholds, though a single tree remains prone to overfitting.
- For mc, Gradient Boosting gave the best fit (R2 = 0.695, RMSE = 2.655), attributed to its sequential error-correction mechanism, which captures nonlinear dependencies between SPT-derived predictors and soil properties.

METHODOLOGICAL NOTE ON DEPLOYMENT (COHESION)
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- Decision Tree achieved the highest independent test-set R² for cohesion and was therefore, whereas Gradient Boosting achieved higher Group CV R² (0.938 vs 0.927), lower fold-to-fold variability, and lower expected overfitting risk.retained as the benchmark model for paper reporting consistency. However, the deployed prediction tool uses Gradient Boosting for cohesion because it demonstrated superior borehole-group cross-validation performance, lower variability across folds, and a lower expected risk of overfitting during future field prediction.

LIMITATION
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- All performance metrics are based on a group-aware train-test split per target; results should be treated as indicative.
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